

Stuyvesant Faculty Perspectives:
Mr. Patrick Honner:
Mathematics is Like a Great Book that Never Ends

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Mr. Patrick Honner has been teaching mathematics for more than 20 years, and he has spent the past 17 years at New York City's specialized high schools, including Stuyvesant since 2019. He holds a B.S. in mathematics and a B.A. in philosophy from Wayne State University, an M.A. in mathematics from the University of Wisconsin, and an M.S.Ed. in mathematics education from the City College of New York.

Honner has received several teaching honors, including the Presidential Award for Excellence in Mathematics and Science Teaching, the Muller Prize for Professional Influence in Education, the University of Chicago's Outstanding Educator Award, and the Math for America's Master Teacher Fellowship (four times). He also won the Sloan Award for Excellence in Teaching Science and Mathematics and was runner-up for the inaugural Rosenthal Prize for Innovation in Math Teaching. Moreover, Honner is a National STEM Ambassador and represents New York City in the New York State Master Teacher Program.

1 Math Without Limits

So, where did it start? Honner says that mathematics has always been part of who he is, a part of how he sees and understands the world. It has always made sense to him. Although he never expected to become a math teacher, once he started learning with, and from, so many brilliant, fun, and creative people, he realized the classroom was where he belonged.

Honner describes math as “a great book that never ends.” He goes on to explain that when you're reading a book, it is inevitable for the story to be over, and you'll have to say goodbye. But there is no end to math, since one keeps discovering new ideas, rediscovering old ones, being challenged, and overcoming those challenges. He remarks that there's so much enlightenment and

satisfaction to be found in math, and there's no limit to how much it can give.

Honner firmly believes that the key to studying mathematics is patience. Some ideas, especially big and important ones, take a long time to truly comprehend. He details that you must learn to live with that frustration, with that sense of incompleteness. Ask any mathematician, and they'll tell you that they spend a lot of their time in a state of uncertainty and confusion. But keep at it, he says, and keep revisiting those big ideas over and over again. In time, things connect and open up.

2 Trying to Do More for Students

Honner finds that few things are as fun and as exciting as learning, and as a teacher, he gets to be around learning every day. Not only is he present when eyes and minds light up with the discovery of new ideas and the mastery of complex techniques, but he also gets to constantly learn alongside everyone else. He feels very fortunate to be able to organize his life around such enjoyable and fulfilling pursuits.

“One of the most challenging aspects of teaching is that you know you could always be doing more for your students,” he says. He explains that this sense is especially true at a place like Stuyvesant, where students are so capable and driven. As a teacher, he knows he could always be pushing his top students a little more, or spending more time with those who need a little extra help, or revising and improving his lessons and activities. He knows that whatever time and energy he puts in will have an impact, but there's only so much time and energy to give.

Honner asserts that great students make great schools. He appreciates the Stuyvesant community, made up of students who are brilliant, creative, fun, curious, driven, and more — as great students everywhere are. But the one thing that really stands out to him is how the students at Stuy truly feel empowered and compelled to lead in the school community. He sees these characteristics in the student organizations and clubs, but also in the hallways and classrooms.

3 Beyond the Classroom

Alongside teaching mathematics, Honner is an active writer and presenter. His work has appeared in *Wired* and the *New York Times* Learning Network, and he writes a column for *Quanta Magazine* where he connects active math research to classrooms and curricula. Honner has authored *Painless Statistics* (2022) and co-authored *SHSAT: New York City Specialized High Schools Admissions Test* (2019). He explains, “I always have a different writing project in the works.”

Honner has hosted events like TEDxNYED, delivered keynotes, and spoken before members of Congress and the Senate. He has given talks on a wide variety of topics, including pure mathematics, teaching and technology, mathematics and creativity, teacher leadership and collaboration, and mathematics and writing. Additionally, Honner conducts workshops for teachers and students, presents at local, national, and international conferences, and participates in mathematical outreach.

4 Looking into the Future

Honner is excited about the formalization movement in mathematics. He provides this example: Mathematicians and computer scientists are essentially developing programming languages that can take a mathematical proof as input, and decide its validity by checking it against the axioms. It is already making headway in a variety of fields, and the possibilities seem pretty incredible. Could an AI math engine someday discover entirely new proofs, or even new theorems? Can you imagine a computer conceiving of and proving a mathematical theorem that humans can't understand? We certainly aren't there yet, but it's exciting to think about, he says.

Honner advises passionate math students to keep taking mathematics courses for as long as they can and keep trying new things. You never know where or when some idea will be useful: linear algebra turns out to be important in machine learning, topology in data science, group theory in chemistry, differential equations in finance, and the list goes on and on and on. He remarks that if you keep learning new things, you'll inevitably see them pop up in whatever it is you're trying to understand. "And surround yourself with good people," he concludes. "And if you aren't surrounded by good people, find good people, and go where they are."